Lecture 6 <u>poll</u>

start with an example

$$\dot{X} = AX \quad , \qquad A = \begin{pmatrix} -3 & 2 \\ 1 & -2 \end{pmatrix}$$

- **Find eigenvalues** of A

$$\operatorname{Det}\begin{pmatrix} -3-\lambda & 2 \\ 1 & -2-\lambda \end{pmatrix} = (-3-\lambda)(-2-\lambda) - 2$$

$$= \lambda^2 + 5\lambda + 4 = (\lambda+4)(\lambda+1) \Rightarrow \lambda = -4, \; \lambda = -1$$

stable node

- **Find eigenvectors** $(A - \lambda I)X = 0$

$\lambda_1 = -4$:
$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} X = \begin{pmatrix} x_1 + 2x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

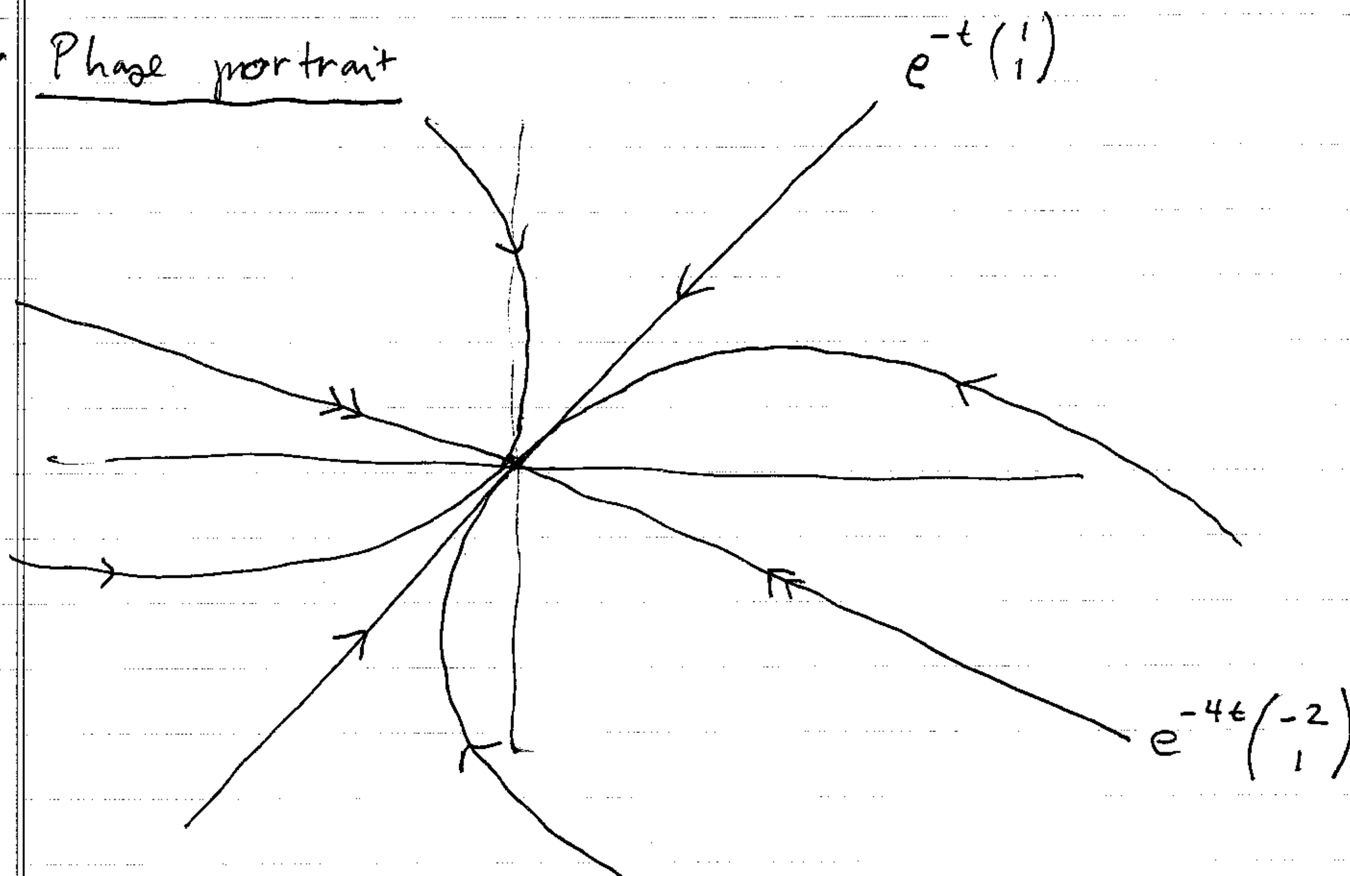
$\lambda_2 = -1$
$$\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} X = \begin{pmatrix} -x_1 + 2x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- **General soln.:**

$$X = c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-4t}\begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

- <u>Phase portrait</u>



$$X(t) = e^{-t}\underbrace{\left(c_1\begin{pmatrix}1\\1\end{pmatrix} + e^{-3t}\begin{pmatrix}-2\\1\end{pmatrix}\right)}$$

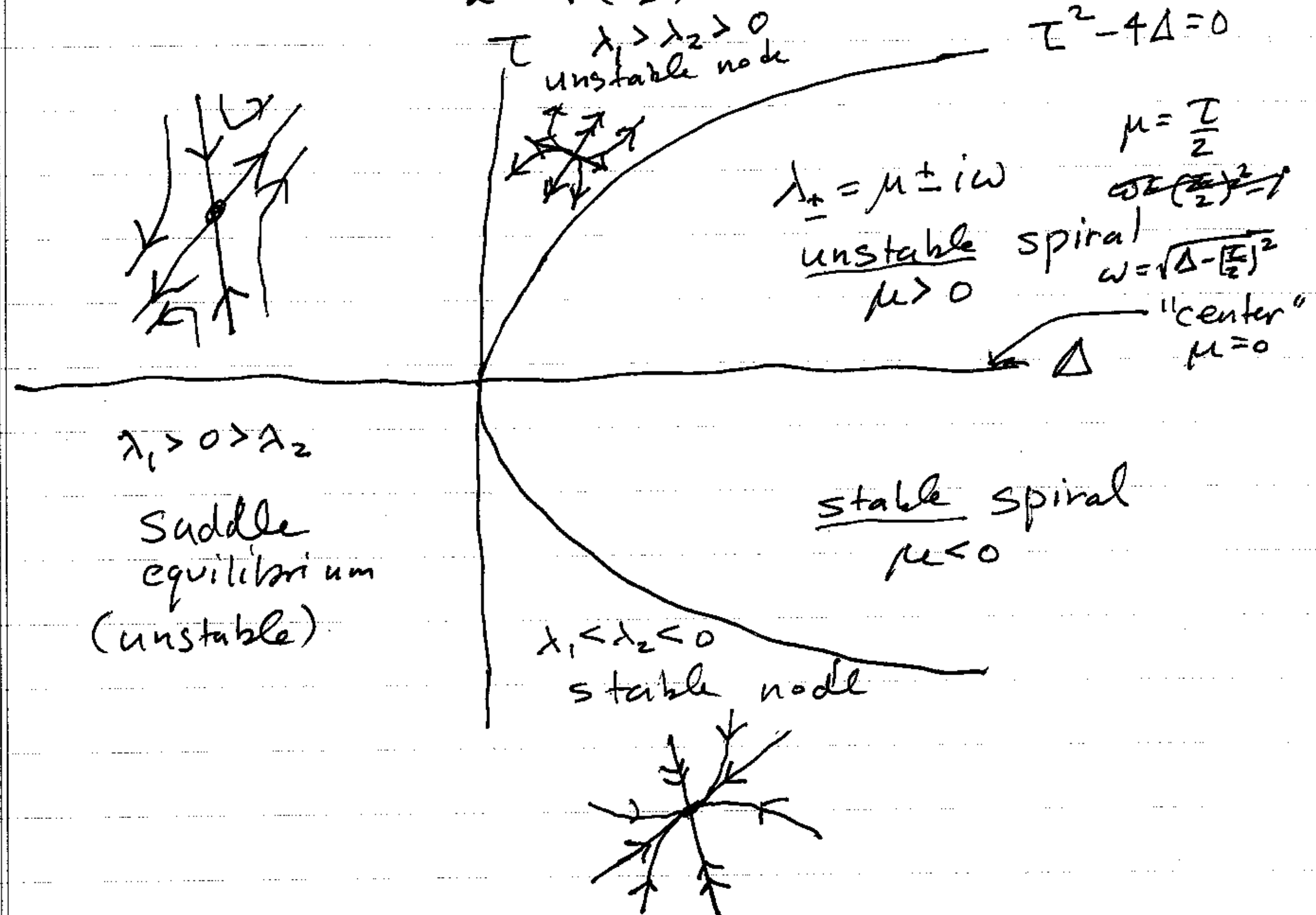
$t \to \infty$ $\quad c_1\begin{pmatrix}1\\1\end{pmatrix}$ dominates

Solns approach origin tangent to "slow direction"

If it were an unstable node, then trajectories leave tangent to "slow direction" e.g. just reverse arrows $t \to -t$

$\dot{X} = AX$, $\quad \tau = \text{Trace}(A)$, $\quad \Delta = \text{Det}(A)$

$$\lambda_\pm = \frac{\tau}{2} \pm \sqrt{\left(\frac{\tau}{2}\right)^2 - \Delta}$$



complex eigenvalue case:

$$X = c\, e^{(\mu + i\omega)t}\, v + c^* e^{(\mu - i\omega)t}\, v^* \quad \leftarrow \text{real } (X = X^*)$$

~~eigenvec~~ everything is complex

$$c = a + ib$$
$$c^* = a - ib$$

$$e^{(\mu + i\omega)t} = e^{\mu t} e^{i\omega t} = e^{\mu t}(\cos(\omega t) + i \sin(\omega t))$$

example with complex eigenvalues

$$\left.\begin{aligned}\dot{x} &= \alpha x - y\\ \dot{y} &= \alpha y + x\end{aligned}\right\} \qquad A = \begin{pmatrix} \alpha & -1 \\ 1 & \alpha \end{pmatrix}$$

eigenvalues $\alpha \pm i$

$$\begin{pmatrix} \alpha-(\alpha\pm i) & -1 \\ 1 & \alpha-(\alpha\pm i) \end{pmatrix} = \begin{pmatrix} \mp i & -1 \\ 1 & \mp i \end{pmatrix}$$

$$\mp i\, x - y = 0 \quad \Rightarrow \quad y = \mp i\, x$$

$$\alpha + i \Rightarrow \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad ; \quad \alpha - i \Rightarrow \begin{pmatrix} 1 \\ i \end{pmatrix}$$

General sol'n:

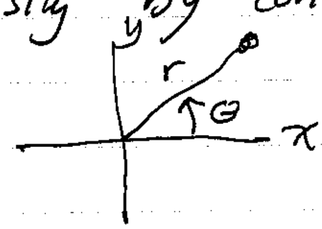
$$X = c\, e^{(\alpha+i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} + c^* e^{(\alpha-i)t} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

let $c = \frac{1}{2}(a+ib)$, $c^* = \frac{1}{2}(a-ib)$, $a, b \in \mathbb{R}$

$$\cdots \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = a e^{\alpha t} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + b e^{\alpha t} \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$$

$\alpha > 0 \Rightarrow$ spiraling out
$\alpha = 0 \Rightarrow$ circular
$\alpha < 0 \Rightarrow$ spiraling in

For this example, we can actually get phase portrait more easily by converting to polar coordinates



$$\left.\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \end{aligned}\right\} \qquad \begin{aligned} x^2+y^2 &= r^2 \\ \frac{y}{x} &= \tan\theta \end{aligned}$$

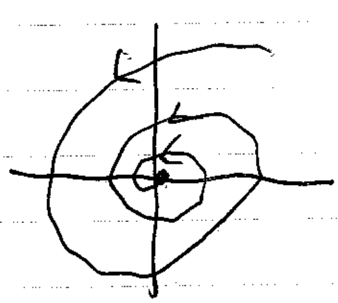
$$\dot{x} = \alpha x - y$$
$$\dot{y} = \alpha y + x$$

$$\dot{x} = \alpha r\cos\theta - r\sin\theta$$
$$\dot{y} = \alpha r\sin\theta + r\cos\theta$$

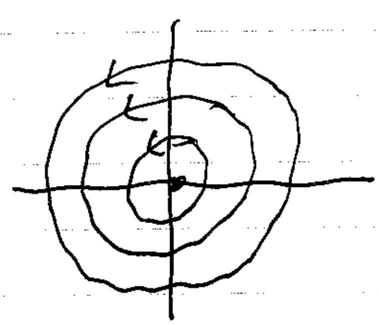
$$\frac{d}{dt}\left(r^2 = x^2+y^2\right)$$
$$2r\dot{r} = 2x\dot{x} + 2y\dot{y}$$
$$\dot{r} = \frac{x}{r}\dot{x} + \frac{y}{r}\dot{y} = \cos\theta\,\dot{x} + \sin\theta\,\dot{y}$$

$$\frac{d}{dt}\left(\frac{y}{x} = \tan\theta\right)$$
$$\frac{\dot{y}x - y\dot{x}}{x^2} = \sec^2\theta\,\dot{\theta}$$
$$\dot{\theta} = \frac{1}{r}\left(\cos\theta\,\dot{y} - \sin\theta\,\dot{x}\right)$$

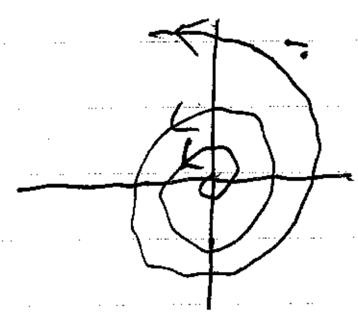
$$\begin{aligned} \dot{r} &= \alpha r \\ \dot{\theta} &= 1 \end{aligned} \Rightarrow \begin{aligned} r(t) &= r_0 e^{\alpha t} \\ \theta(t) &= \theta_0 + t \end{aligned} \Rightarrow \begin{aligned} x(t) &= r_0 e^{\alpha t}\cos(\theta_0+t) \\ y(t) &= r_0 e^{\alpha t}\sin(\theta_0+t) \end{aligned}$$



$\alpha < 0$



$\alpha = 0$



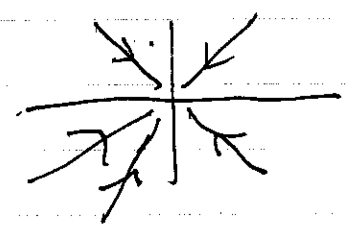
$\alpha > 0$

$\tau^2 = 4\Delta$ Boundary case eigenvalue is $\frac{\tau}{2} = \lambda$, which has multiplicity 2

2 cases

1 eigenvector only (geometric multiplicity = 1)
2 eigenvectors v_1, v_2 that are linearly independent

$$X(t) = e^{\lambda t}\underbrace{(c_1 v_1 + c_2 v_2)}_{X_0} = X_0 e^{\lambda t}$$



$\lambda < 0$ example "star"

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \qquad b \neq 0$$

eigenvectors: $\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} by \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = 0$

only eigenvector is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

need a second soln.

$\dot{x} = \lambda x + by$
$\dot{y} = \lambda y \rightarrow y(t) = y_0 e^{\lambda t}$

$\dot{x} = \lambda x + b y_0 e^{\lambda t}$
$\dot{x} - \lambda x = b y_0 e^{\lambda t}$
$e^{-\lambda t}(\dot{x} - \lambda x) = b y_0$
$\frac{d}{dt}(e^{-\lambda t} x) = b y_0$

$$e^{-\lambda t}x - x_0 = b y_0 t$$

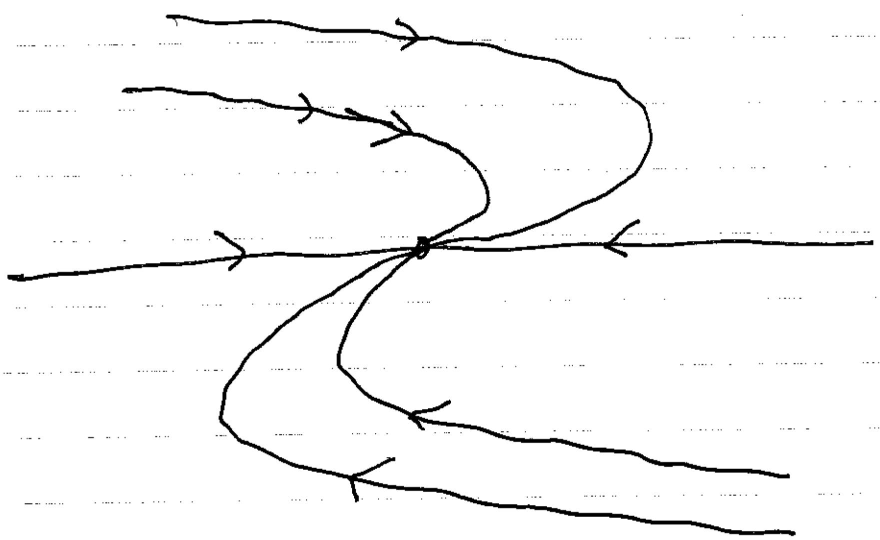
$$x = x_0 e^{\lambda t} + b y_0 t e^{\lambda t}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = x_0 \begin{pmatrix} e^{\lambda t} \\ 0 \end{pmatrix} + y_0 \begin{pmatrix} b t e^{\lambda t} \\ e^{\lambda t} \end{pmatrix}$$

$$= \cancel{x_0 \begin{pmatrix} e^{\lambda t} \\ 0 \end{pmatrix} + y_0 e^{\lambda t}}$$

$$= x_0 e^{\lambda t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y_0 e^{\lambda t} \begin{pmatrix} b t \\ 1 \end{pmatrix}$$

Typical phase portrait för (stable) degenerate node



$\lambda < 0$
$b > 0$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\left(\begin{array}{c} \text{on } y \text{ axis} \\ \dot{x} = b y \\ \dot{y} = \lambda y \end{array} \right)$$